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DAA LAB

LAB EXPERIMENTS

EXPERIMENT – 7

INDEX OF FIRST OCCURRENCE IN A STRING

AIM:

To find the index of the first occurrence of a given character in a string using Python and analyze its time complexity.

ALGORITHM:

1. Start the program.
2. Initialize a string and a target character.
3. Traverse the string from left to right.
4. Compare each character with the target character.
5. If a match is found, display its index and stop searching.
6. If no match is found, display -1.
7. Stop the program.

PROCEDURE:

1. Declare a sorted array and the target element.
2. Declare a string and the character to be searched.
3. Iterate through the string using a loop.
4. Compare each character with the target.
5. Store the index when the first match occurs.
6. Print the index and time complexity.

PYTHON CODE:

```python
s = "programming"
target = "g"
index = -1
for i in range(len(s)):
    if s[i] == target:
        index = i
        break
print("String:", s)
print("Target character:", target)
print("First occurrence index:", index)
print("Time Complexity: O(n)")
```

SAMPLE OUTPUT:

```
Output

String: programming
Target character: g
First occurrence index: 3
Time Complexity: O(n)
```

RESULT:

The first occurrence of the target character was found successfully at index 3. The worst-case time complexity is (O(n)), and the space complexity is (O(1)).